HA-4

1) using generic Algorithm maximize the function MAX $f(x) = x^2 - 4x + 4$ in interval $[0, 16]$

Sol:-

Given $f(x) = x^2 - 4x + 4$, where $x \in [0, 16]$

Since $f(x) = (x-2)^2$ this parabola.

Equation:-

$f(0) = (0-2)^2 = 4$ $\quad$ $f(1) = (1-2)^2 = 1$

$f(2) = (2-2)^2 = 0$ $\quad$ $f(3) = (3-1)^2 = 1$

$f(4) = (4-2)^2 = 4$ $\quad$ $f(5) = (5-3)^2 = 9$

$f(6) = (6-2)^2 = 16$ $\quad$ $f(7) = (7-2)^2 = 25$

$f(8) = (8-2)^2 = 36$ $\quad$ $f(9) = (9-2)^2 = 49$

$f(10) = (10-2)^2 = 64$ $\quad$ $f(11) = (11-2)^2 = 81$

$f(12) = (12-2)^2 = 100$ $\quad$ $f(13) = (13-2)^2 = 121$

$f(14) = (14-2)^2 = 144$ $\quad$ $f(15) = (15-12)^2 = 169$

$f(16) = (16-2)^2 = 196$

$\therefore$ Maximum Value $f(16) = 196$

2) Given $f(x) = -x^2 + 3x + 10$.

Intial Positions

$x_1 = -9.6$ $x_2 = -9$, $x_3 = -4.6$, $x_4 = -2.1$, $x_5 = 1.6$

$x_6 = 3.3$, $x_7 = 5.8$, $x_8 = 9.3$

$f(x_1) = -(-9.6)^2 + 3(-9.6) + 10 = -110.96$

$f(x_2) = -(-9)^2 + 3(-9) + 10 = -98$

$f(x_3) = -(-4.6)^2 + 3(-4.6) + 10 = -24.96$

$f(x_4) = -(-2.1)^2 + 3(-2.1) + 10 = -0.71$

$f(x_5) = -(1.6)^2 + 3(1.6) + 10 = 12.24$

$f(x_6) = -(3)^2 + 3(3) + 10 = 9.01$

$f(x_7) = -(5.8)^2 + 3(5.8) + 10 = -6.24$

$x_{best} = x_5 = 1.6$

$f(x_{best}) = 12.24$

velocity

$v_1 = c_1 r_1 (p_{best} - x_i) + c_2 r_2 (g_{best} - x_i)$

$P_{best} = x_i r_i = 0$

$v_1 = 0.816 (1.6 + 9.6) = 9.139$

$x_2$ $v_2 = 0.816(10.6) = 8.650$

$x_2' = -9 + 8.650 = -0.35$

$x_3$

$v_3 = 0.816(6.2) = 5.059$

$x_3' = -4.6 + 5.059 = 0.459$.

$x_4$

$v_4 = 0.816(8.7) = 3.009$

$x_4' = -2.1 + 3.0192 + 0.919$

$x_5$ $v_5 = 0$

$x_5' = 1.6$.

$f(2.373) = -5.63 + 7.12 + 10$
$= 12.49$

$x_{best} = 2.73$

$f(x_{best}) = 12.49$

3) Distance matrix

| $C_1$ | $C_2$ | $C_3$ | $C_4$ |
|---|---|---|---|
| 0 | 4 | 15 | 1 |
| 4 | 0 | 5 | 8 |
| 15 | 5 | 0 | 4 |
| 1 | 8 | 4 | 0. |

Number of cities = 4

Heuristic value $x_{ij} = 1/d_{ij}$

2. Intial position $C_1$

Possible moves

$C_1 \to C_2$ distance = 4

$C_1 \to C_3$ distance = 15

$C_1 \to C_4$ distance = 1 ✓

3) $C_1 \to C_4$

Possible moves

$C_4 \to C_2$ distance = 8

$C_4 \to C_3$ distance = 4 ✓

$C_1 \to C_4 \to C_3$

4) $C_1 \to C_4 \to C_3$

Possible moves

$C_3 \to C_2$ distance = 5

$\therefore C_1 \to C_4 \to C_3 \to C_2 \to C_1$

Total cost = 1+4+5+4 = 14

4Q) Given

$f(x) = x^2 + 8x$

$0 \le x \le 24$

Intial population

"00110", "01001", "00110", "10010"

00110 → $x_1 = 6$

01001 → $x_2 = 9$

01110 → $x_3 = 14$

10010 → $x_4 = 18$

$f(x) = x^2 + 8x$

$f(x_1) = 36 + 48 = 84$

$f(x_2) = 81 + 72 = 153$

$f(x_3) = 196 + 112 = 308$

$f(x_4) = 324 + 144 = 468$.

Best $x_4 = 18$.

Parents

16 → 10010

14 → 01110

After Consumer

10010 → 22

01010 → 10

New population

22, 11, 18, 14.

New fitness

$f(22) = 484 + 196 = 660$

$f(11) = 121 + 88 = 209$

$f(18) = 468$.

$f(14) = 308$

$x = 22$

Population moves towards maximum $x = 24$

$x_{best} = 24$

$f(x_{best}) = 768$

5) use Simulated Algorithm method to store the following maximite for $f(x) = 2x_1 + x_2^2$

$0 \le x_1, x_2 \le 9$

Sol:-

given

$f(x) = 2x_1 + x_2^2$

$0 \le x_1, x_2 \le 9$

1. $x_1 = 6, x_2 = 6$.

$f(6,6) = 2(6) + 6^2$

$= 12 + 36 = 48$

$T = 10$

2. $x_1 = 5, x_2 = 5$

$f(5,5) = 10 + 25 = 35$

$\Delta f = 35 - 48 = -13$

3) $x_1 = 4$ , $x_2 = 4$

$f(4,4) = 8 + 16 + 24$

$\Delta f = 24 - 35 = -11$

4. $x_1 = 3$ , $x_2 = 3$

$f(3,3) = 6 + 9 = 15$

$\Delta f = 15 - 24 = -9$

5. $x_1 = 2$ , $x_2 = 2$ .

$f(2,2) = 4 + 4 = 8$ .

6. $x_1 = 1$ , $x_2 = 1$

$f(1,1) = 2 + 1 = 3$

Reduce Impedence

$T = 10 \to 5 \to 1$

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